

Run report

2026-09-11T160419Z

Configuration

Parameter	Value	Parameter	Value
batch_size	1,024	n_samples	1,600,000
dim	12	dataset	jets
seed	1,851,207,501	data_seed	42
hidden_units	128	n_layers	3
n_epochs	100	n_disc_steps	5
lr_g	3.000×10^{-5}	lr_d	1.000×10^{-4}
lambda_dispersion	0.01500	mmd_subsample	16,384
mmd_test	2.573×10^{-4}	best_epoch	91
compile_cache_warm	True		
variables	m [GeV], $\ln \rho$, $\lambda_{0.5}^1$, w , λ_2^1 , z_g , $\tau_{21}^{(\beta=1)}$, M , n_{ch} , f_{ch} , p_T^D , q		
mmd_sigmas_detector	$3.107 \times (\frac{1}{2}, \frac{1}{\sqrt{2}}, 1, \sqrt{2}, 2)$		
mmd_sigmas_particle	$3.027 \times (\frac{1}{2}, \frac{1}{\sqrt{2}}, 1, \sqrt{2}, 2)$		

Timing

Phase	Seconds	Share (%)	Detail
train	40.643	57.9	train
transfer	5.714		train
compile	3.910		train
epochs	29.729		train
select	1.080		train
particle_mmd	0.403	0.6	train
save	0.352	0.5	train
data	16.574	23.6	cache hit – load
load	1.756	2.5	load
plots	10.416	14.8	load
evaluate	0.008	0.0	load
total	70.152	100.0	

Detector-level metrics

Distance to Data. Improvements are relative to Simulation.

Wasserstein Distance [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
Mass and hard scale (IRC-safe kinematics)							
m [GeV]	208.9	138.8	33.6	9.648	95.4	9.340	95.5
$\ln \rho$	227.9	20.71	90.9	23.16	89.8	11.60	94.9
Continuous angularities (IRC-safe jet shapes)							
$\lambda_{0.5}^1$	262.6	12.84	95.1	18.66	92.9	7.440	97.2
w	265.6	9.978	96.2	16.75	93.7	6.082	97.7
λ_2^1	255.4	9.215	96.4	15.26	94.0	6.440	97.5
Splitting and 2-prong substructure							
$z_g^\dagger$	60.45	60.45	0.0	15.93	73.6	17.11	71.7
$\tau_{21}^{(\beta=1)}$	121.8	70.10	42.4	37.71	69.0	18.78	84.6
Hadronization, multiplicity and fragmentation (IRC-unsafe)							
M	224.2	177.3	20.9	12.98	94.2	20.78	90.7
n_{ch}	220.8	172.3	22.0	13.59	93.8	22.69	89.7
f_{ch}	52.86	40.94	22.6	5.765	89.1	7.496	85.8
p_T^D	256.8	30.80	88.0	14.45	94.4	16.82	93.4
q	79.67	17.51	78.0	3.901	95.1	5.595	93.0

[†] IBU's purity binning produced a single bin, so IBU failed to unfold.

JS divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
Mass and hard scale (IRC-safe kinematics)							
m [GeV]	6.762	3.666	45.8	0.07863	98.8	0.06873	99.0
$\ln \rho$	8.383	0.4017	95.2	0.2312	97.2	0.1500	98.2
Continuous angularities (IRC-safe jet shapes)							
$\lambda_{0.5}^1$	9.359	0.4714	95.0	0.1792	98.1	0.1141	98.8
w	9.139	0.4527	95.0	0.1621	98.2	0.1155	98.7
λ_2^1	7.776	0.1728	97.8	0.1488	98.1	0.1204	98.5
Splitting and 2-prong substructure							
$z_g^\dagger$	0.6723	0.6723	0.0	0.1112	83.5	0.1136	83.1
$\tau_{21}^{(\beta=1)}$	3.372	1.934	42.7	0.3410	89.9	0.1702	95.0
Hadronization, multiplicity and fragmentation (IRC-unsafe)							
M	9.640	7.633	20.8	0.1401	98.5	0.1704	98.2
n_{ch}	7.512	6.314	16.0	0.1049	98.6	0.1431	98.1
f_{ch}	1.033	0.7960	22.9	0.08459	91.8	0.09115	91.2
p_T^D	8.223	1.311	84.1	0.1142	98.6	0.1381	98.3
q	1.576	0.2512	84.1	0.07234	95.4	0.08142	94.8

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VLC divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
Mass and hard scale (IRC-safe kinematics)							
m [GeV]	26.93	14.62	45.7	0.2934	98.9	0.2578	99.0
$\ln \rho$	33.31	1.576	95.3	0.9044	97.3	0.5797	98.3
Continuous angularities (IRC-safe jet shapes)							
$\lambda_{0.5}^1$	37.19	1.877	95.0	0.7107	98.1	0.4506	98.8
w	36.32	1.800	95.0	0.6406	98.2	0.4541	98.7
λ_2^1	30.90	0.6583	97.9	0.5567	98.2	0.4426	98.6
Splitting and 2-prong substructure							
$z_g^\dagger$	2.683	2.683	0.0	0.4446	83.4	0.4545	83.1
$\tau_{21}^{(\beta=1)}$	13.44	7.712	42.6	1.359	89.9	0.6761	95.0
Hadronization, multiplicity and fragmentation (IRC-unsafe)							
M	38.25	30.31	20.8	0.5292	98.6	0.6497	98.3
n_{ch}	29.81	25.07	15.9	0.3928	98.7	0.5460	98.2
f_{ch}	4.125	3.177	23.0	0.3383	91.8	0.3645	91.2
p_T^D	32.65	5.211	84.0	0.4522	98.6	0.5479	98.3
q	6.261	0.9940	84.1	0.2796	95.5	0.3147	95.0

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Particle-level metrics

Distance to Truth. Improvements are relative to Generation.

Wasserstein [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
Mass and hard scale (IRC-safe kinematics)							
m [GeV]	111.7	94.07	15.8	89.36	20.0	86.32	22.8
$\ln \rho$	253.1	32.57	87.1	13.26	94.8	17.29	93.2
Continuous angularities (IRC-safe jet shapes)							
$\lambda_{0.5}^1$	303.0	18.71	93.8	16.66	94.5	6.176	98.0
w	284.1	12.29	95.7	12.51	95.6	3.450	98.8
λ_2^1	253.8	11.13	95.6	9.086	96.4	3.172	98.8
Splitting and 2-prong substructure							
$z_g^\dagger$	113.4	113.4	0.0	25.05	77.9	24.69	78.2
$\tau_{21}^{(\beta=1)}$	259.2	150.5	41.9	45.47	82.5	11.62	95.5
Hadronization, multiplicity and fragmentation (IRC-unsafe)							
M	256.8	204.8	20.2	30.94	88.0	24.19	90.6
n_{ch}	234.6	188.6	19.6	32.00	86.4	24.70	89.5
f_{ch}	45.78	45.33	1.0	4.212	90.8	5.079	88.9
p_T^D	236.3	55.36	76.6	16.46	93.0	17.61	92.5
q	79.37	18.15	77.1	3.499	95.6	5.307	93.3

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JS divergence [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
Mass and hard scale (IRC-safe kinematics)							
m [GeV]	1.963	11.86	−504.3	1.650	16.0	1.596	18.7
$\ln \rho$	9.878	1.731	82.5	0.2391	97.6	0.1995	98.0
Continuous angularities (IRC-safe jet shapes)							
$\lambda_{0.5}^1$	12.34	2.009	83.7	0.2387	98.1	0.1552	98.7
w	11.39	1.465	87.1	0.2232	98.0	0.1688	98.5
λ_2^1	8.824	0.8335	90.6	0.1487	98.3	0.1161	98.7
Splitting and 2-prong substructure							
$z_g^\dagger$	2.041	2.041	0.0	0.2012	90.1	0.1972	90.3
$\tau_{21}^{(\beta=1)}$	10.50	6.577	37.4	0.5327	94.9	0.1412	98.7
Hadronization, multiplicity and fragmentation (IRC-unsafe)							
M	7.842	12.83	−63.7	0.2576	96.7	0.2621	96.7
n_{ch}	6.447	9.773	−51.6	0.1882	97.1	0.1872	97.1
f_{ch}	1.025	0.9435	7.9	0.1005	90.2	0.1121	89.1
p_T^D	8.342	4.029	51.7	0.2553	96.9	0.2841	96.6
q	2.027	0.9958	50.9	0.09290	95.4	0.1102	94.6

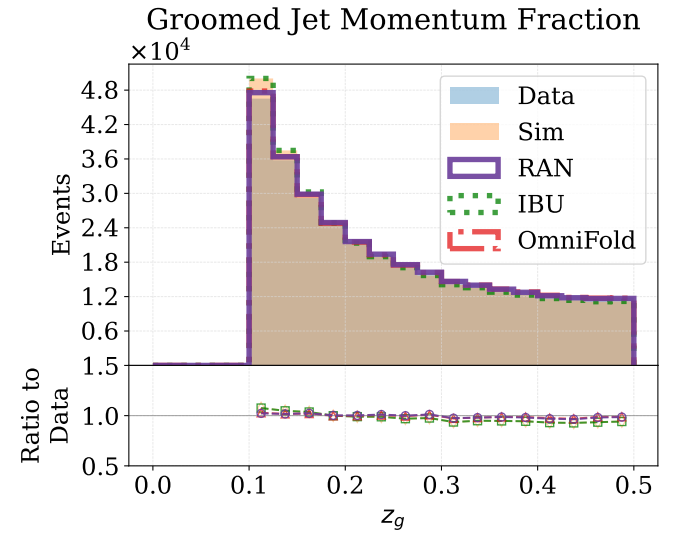
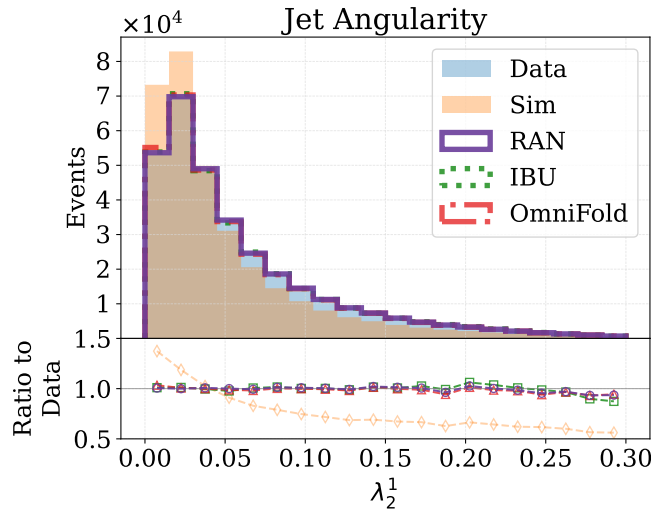
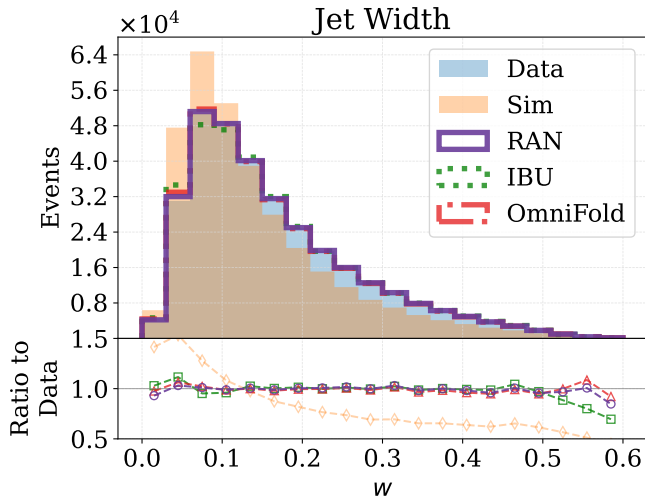
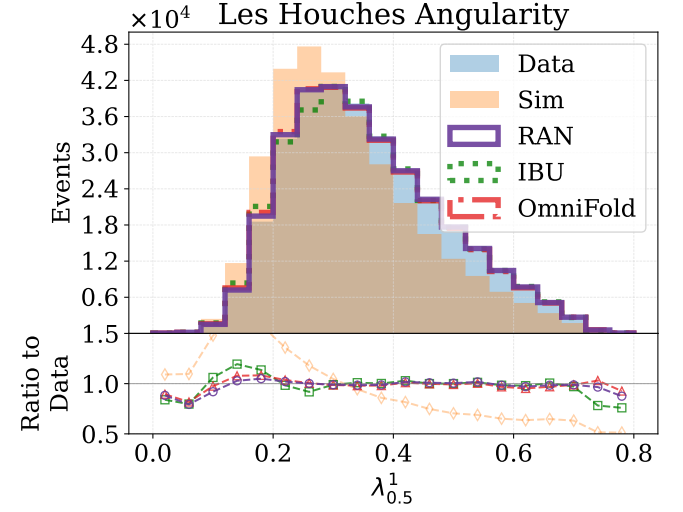
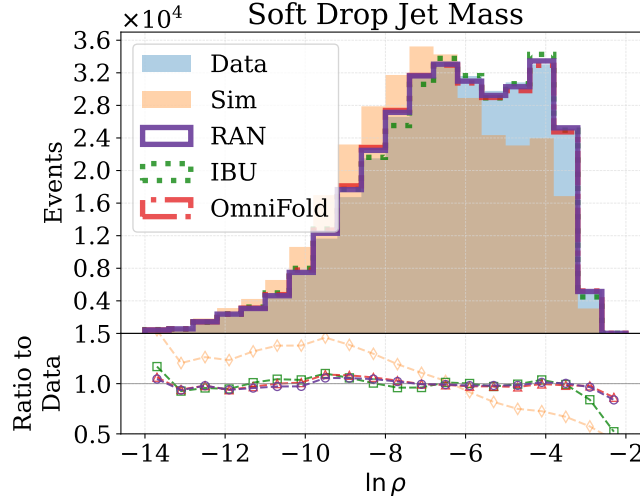
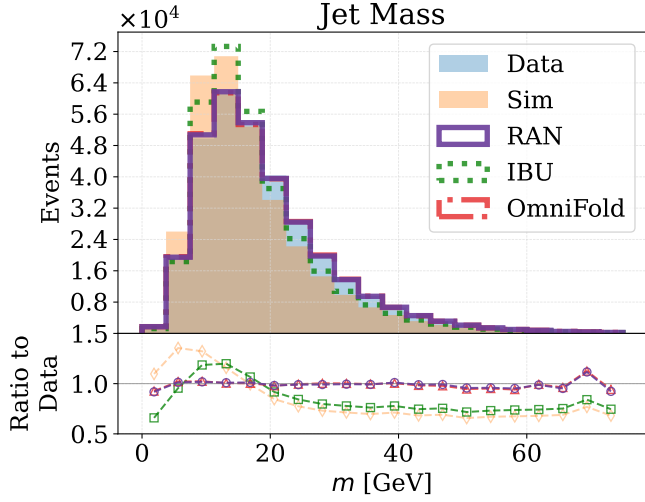
[†] IBU's purity binning produced a single bin, so IBU failed to unfold.

VLC divergence [$\times 10^3$]

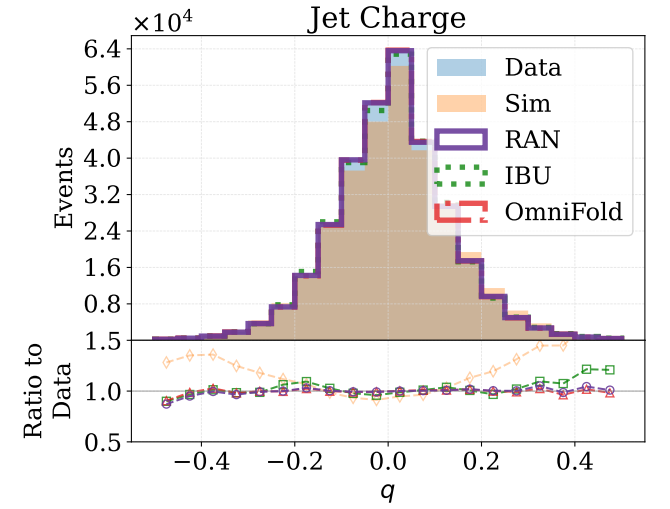
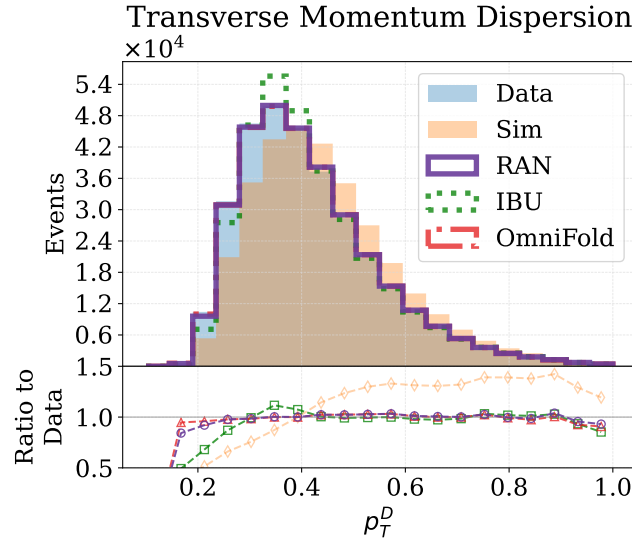
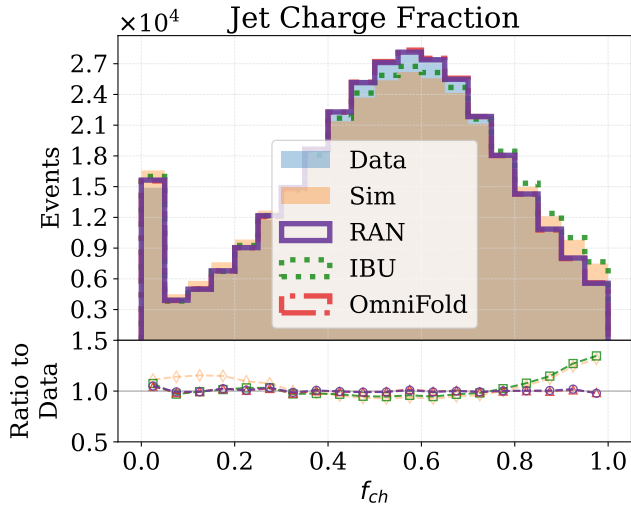
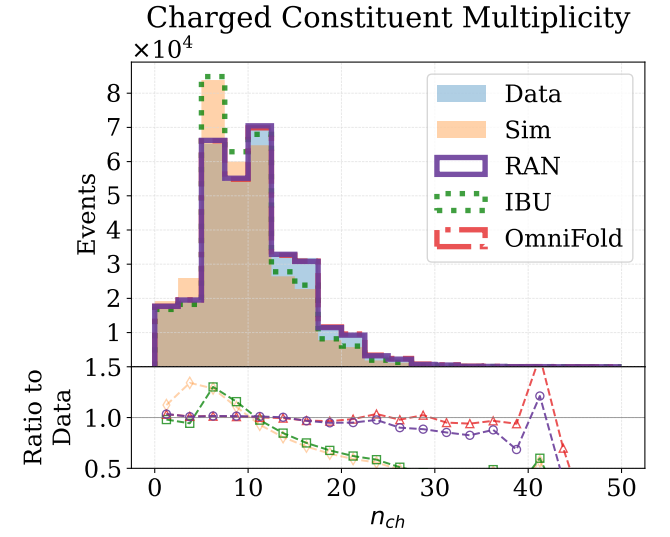
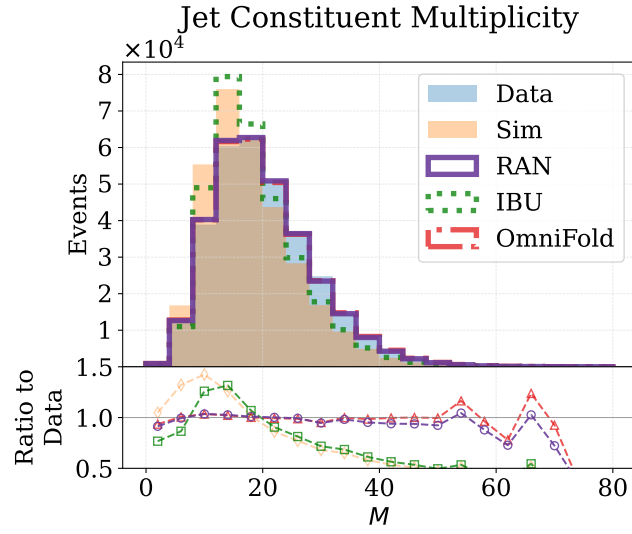
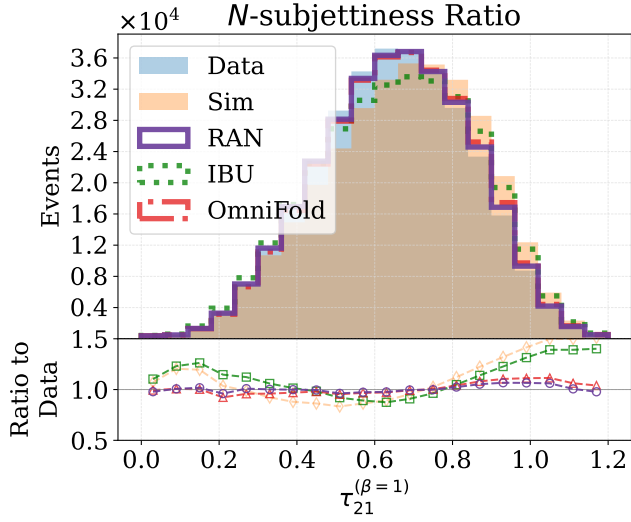
Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
Mass and hard scale (IRC-safe kinematics)							
m [GeV]	7.809	45.93	−488.2	6.555	16.1	6.347	18.7
$\ln \rho$	39.27	6.889	82.5	0.9406	97.6	0.7830	98.0
Continuous angularities (IRC-safe jet shapes)							
$\lambda_{0.5}^1$	48.98	8.014	83.6	0.9484	98.1	0.6153	98.7
w	45.23	5.837	87.1	0.8724	98.1	0.6574	98.5
λ_2^1	35.08	3.315	90.5	0.5746	98.4	0.4475	98.7
Splitting and 2-prong substructure							
$z_g^\dagger$	8.086	8.086	0.0	0.8033	90.1	0.7875	90.3
$\tau_{21}^{(\beta=1)}$	41.75	26.17	37.3	2.123	94.9	0.5580	98.7
Hadronization, multiplicity and fragmentation (IRC-unsafe)							
M	31.10	49.85	−60.3	0.9881	96.8	1.005	96.8
n_{ch}	25.58	38.38	−50.1	0.7160	97.2	0.7119	97.2
f_{ch}	4.095	3.766	8.0	0.4015	90.2	0.4480	89.1
p_T^D	33.05	16.05	51.4	1.010	96.9	1.121	96.6
q	7.982	3.921	50.9	0.3250	95.9	0.3856	95.2

[†] IBU’s purity binning produced a single bin, so IBU failed to unfold.

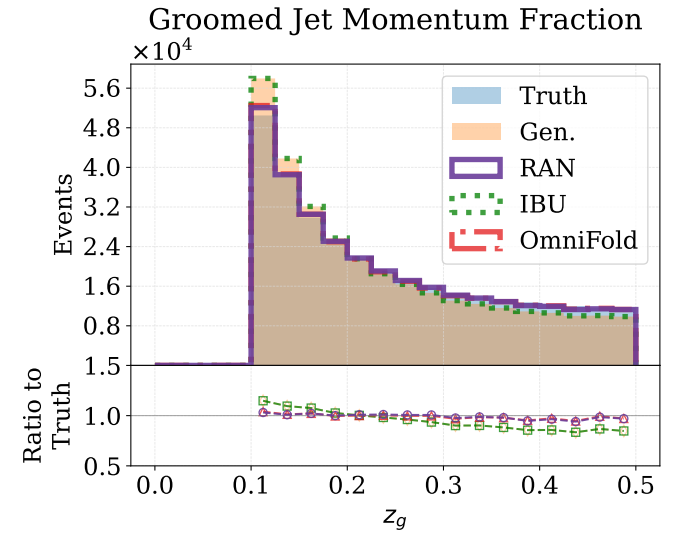
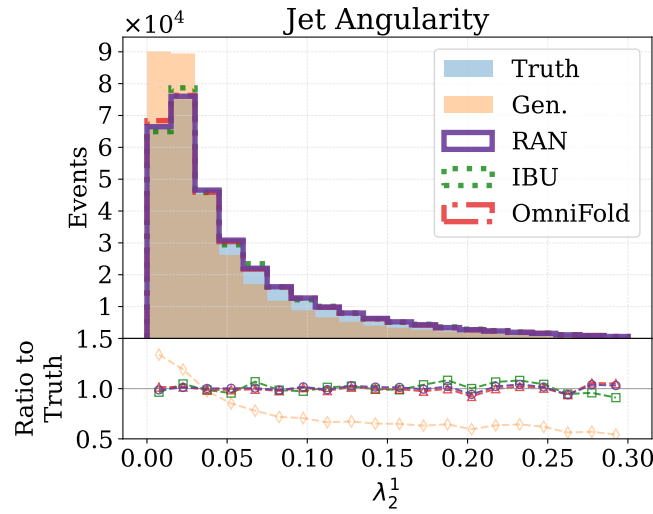
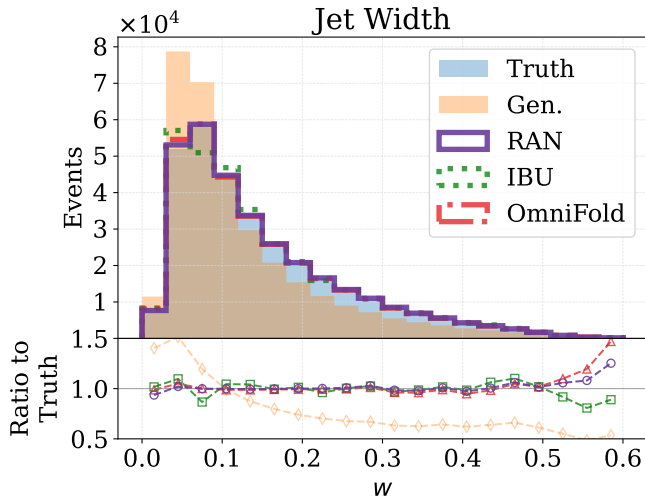
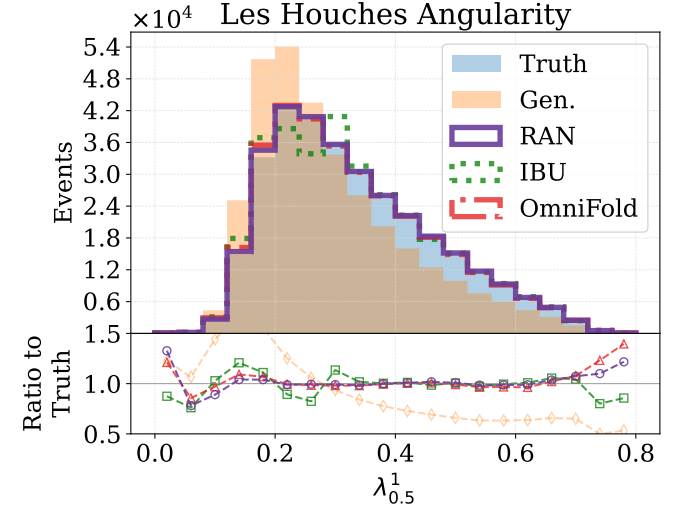
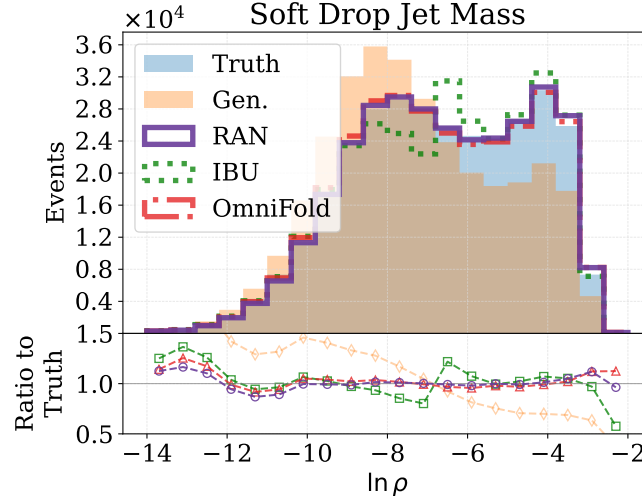
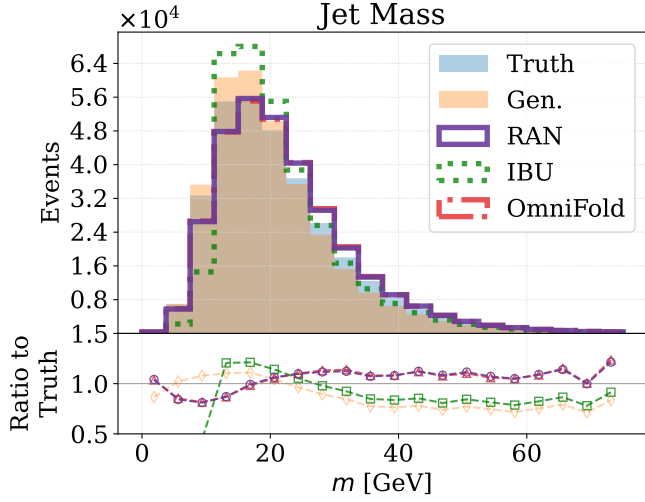
Detector Level (1 of 2)



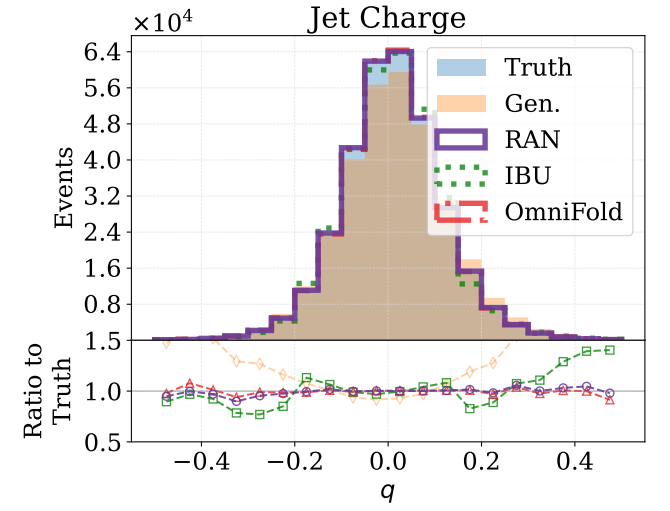
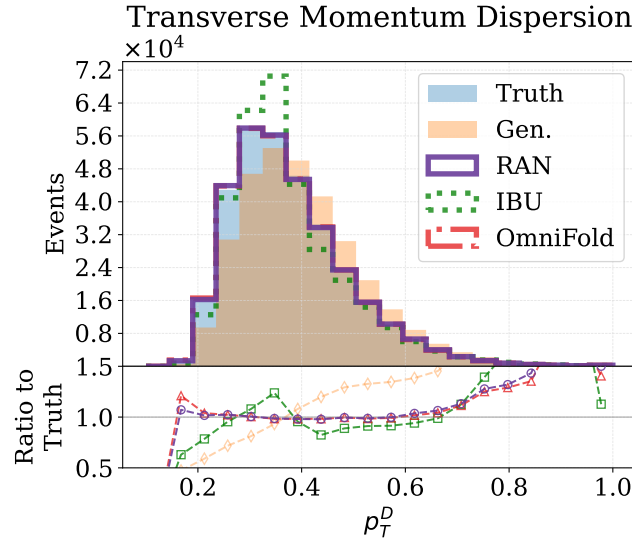
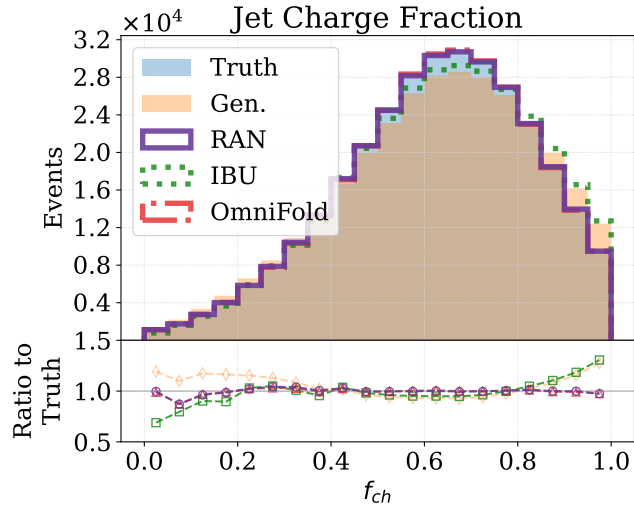
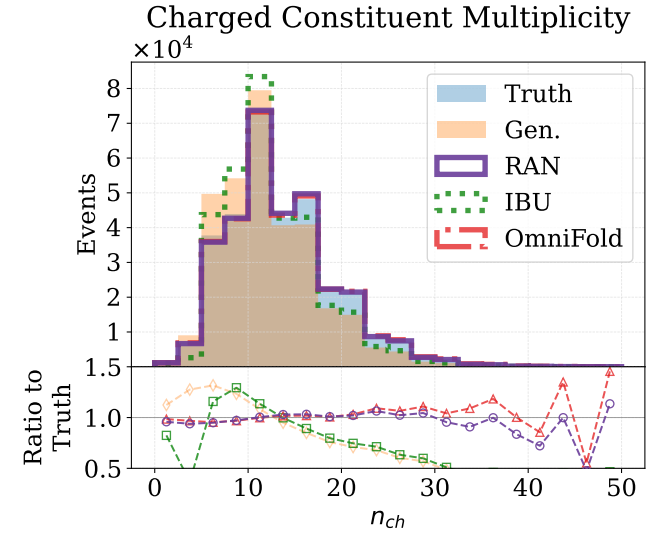
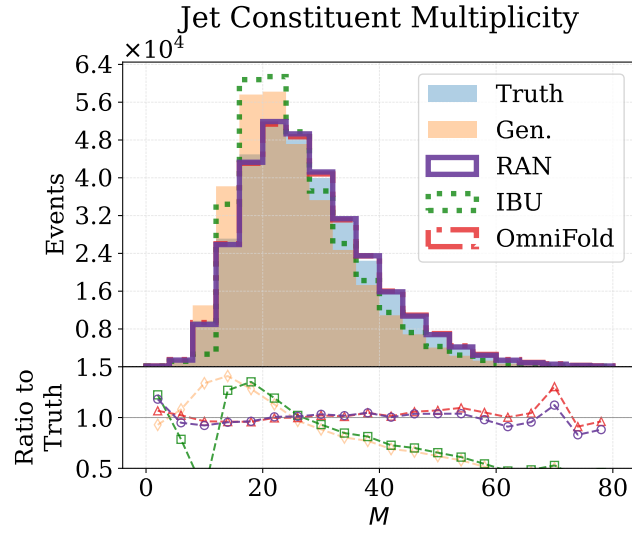
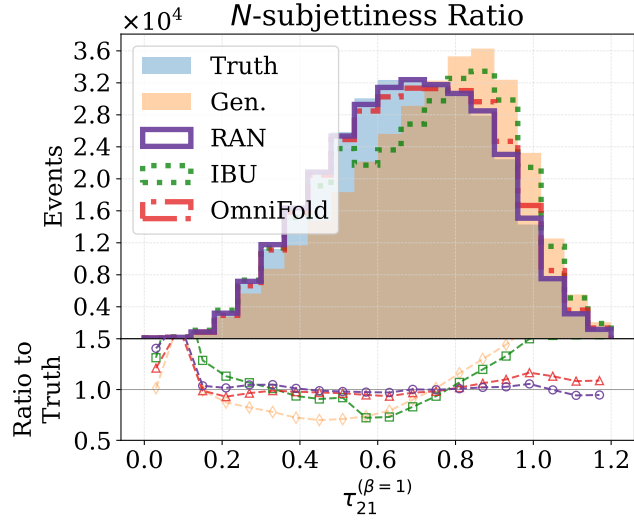
Detector Level (2 of 2)



Particle Level (1 of 2)

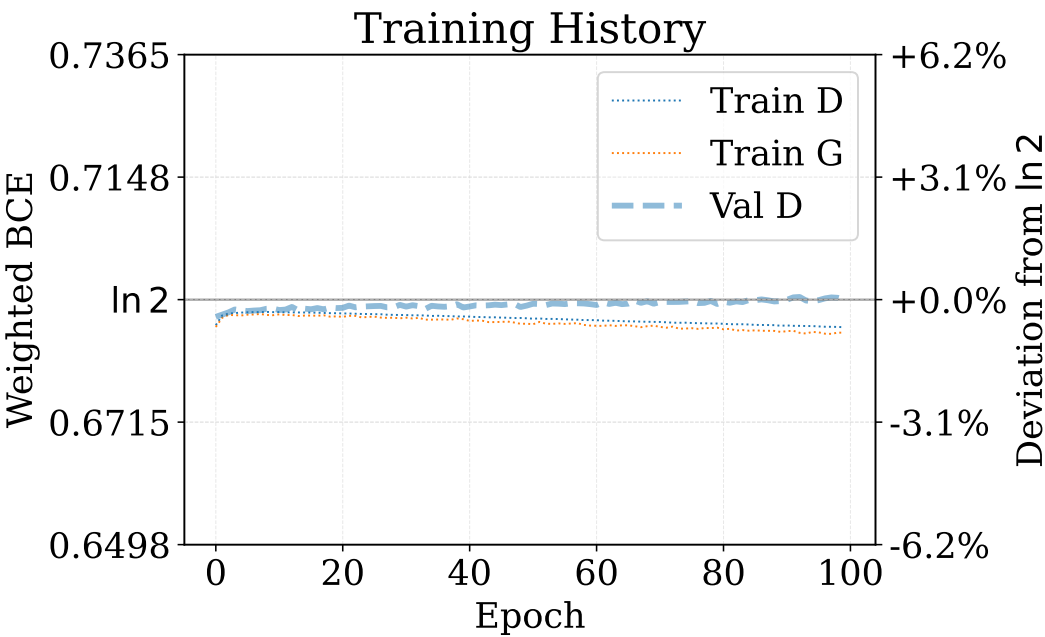


Particle Level (2 of 2)



Training diagnostics

Training losses



Selection criteria

